

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
Belagavi-590018



A PROJECT REPORT
ON
(BEC786)

“ SMART ASSISTIVE SYSTEM FOR ALZHEIMER’S
PATIENTS USING IOT AND MACHINE LEARNING ”

Submitted in the partial fulfilment of the requirements for the Award of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING
BY

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Affiliated to VTU Belgaum and Approved by AICTE, New Delhi, Recognised by Govt. of Karnataka,
Accredited by NAAC with ‘A+’, NBA (ECE,CSE,ISE,ISE,CV,ME,EEE)
Raja Reddy Layout, Chikkabanavara, Bengaluru – 560 090

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CERTIFICATE

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ABSTRACT

The increasing need for intelligent healthcare solutions has motivated the development of a smart assistive system that uses IoT and machine learning to support Alzheimer's patients through continuous monitoring, automated alerts, and intelligent decision-making. This project integrates physiological sensing, motion detection, facial recognition, and cloud connectivity to create a comprehensive, non-intrusive health-monitoring ecosystem. Using ESP32/ESP8266 for real-time data acquisition, multiple biomedical sensors to capture vitals, and Firebase for cloud synchronization, the system monitors heart rate, SpO₂, ECG patterns, temperature, fall events, and patient movement with high accuracy. A camera module combined with Python-based face recognition enables automatic identification of patients, caregivers, and visitors, enhancing safety and preventing unauthorized access. The motivation behind this system lies in addressing the growing challenges in elderly care by offering an affordable, automated, and reliable solution capable of functioning in homes, hospitals, and assisted-living environments. Unlike traditional monitoring that depends solely on manual observation, this system provides continuous, real-time alerts through mobile dashboards, audio announcements, and GSM-based emergency messages, ensuring timely intervention during critical events.

Beyond its technical architecture, the system emphasizes accessibility and caregiver support, reducing workload by automating vital checks, tracking patient location through GPS, and logging behavioral and environmental data for long-term analysis. The interface displays real-time vitals, activity patterns, fall detections, and identity logs, making it easier for caregivers to interpret patient conditions quickly. The overall aim is to deliver a responsive, intelligent, and user-friendly assistive system that advances modern elderly care by combining IoT sensing, machine learning analytics, and real-time monitoring to create a safer and more supportive environment for Alzheimer's patients.

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CHAPTER 1

INTRODUCTION

1.1 Background

Alzheimer's disease is a progressive neurodegenerative disorder that primarily affects memory, cognition, and behaviour. Patients with Alzheimer's often experience disorientation, wandering, and difficulty recognizing familiar faces, leading to significant safety concerns. As the disease advances, individuals lose their ability to perform daily activities, increasing their dependence on caregivers. The burden on caregivers and family members is immense, requiring constant monitoring of the patient's health and location. Existing solutions lack a comprehensive system that integrates real time health monitoring, fall detection, location tracking, and facial recognition.

This project aims to bridge this gap by developing an integrated assistive system to enhance patient safety, independence, and caregiver support. The Smart Assistive System for Alzheimer's Patients is designed to provide real-time health monitoring, fall detection, GPS tracking, and facial recognition to assist both patients and caregivers. The Smart Assistive System for Alzheimer's Patients integrates IoT, AI, and wearable technology to provide real-time health monitoring, facial recognition, and emergency alert Notifications.

1.2 Objectives

- To Develop an integrated IoT-based health monitoring wearable for Alzheimer's patients
- To Implement machine learning algorithms for fall detection with 95%+ accuracy
- To Create real-time location tracking system with GPS integration
- To Develop facial recognition module for cognitive assistance
- To Build mobile application for caregivers with alert notifications
- To Establish secure data transmission and storage protocols
- To Provide cost-effective solution accessible to middle-income families

1.3 Motivation

The motivation for developing this Smart Assistive System stems from several urgent medical, social, and technological challenges faced by Alzheimer's patients, caregivers, and healthcare institutions.

- Patient Safety and Risk Reduction

Alzheimer's disease leads to severe memory impairment, disorientation, and confusion, putting patients at high risk of wandering, falls, accidental injuries, and forgetting essential tasks such as turning off appliances or taking medication. Globally, more than 60% of Alzheimer's patients wander at least once, and many face life-threatening situations due to lack of continuous monitoring. Traditional caregiving methods cannot provide round-the-clock vigilance, especially in home environments. A smart IoT-enabled solution can provide real-time tracking, emergency alerts, hazard detection, and intelligent preventive mechanisms to ensure patient safety.

- Early Detection and Behavioral Analysis

Alzheimer's progression is often unpredictable. Subtle changes in movement patterns, sleep cycles, forgetfulness, or daily routines usually go unnoticed until major decline occurs. Machine learning systems can analyze real-time sensor data, detect anomalies, and identify early warning signs of cognitive or physical deterioration. Such early alerts empower doctors and caregivers to intervene sooner, personalize care plans, and improve the long-term quality of life for patients.

- Medication & Routine Management

One of the biggest challenges for Alzheimer's patients is remembering daily tasks, especially medication schedules, hydration routines, and nutritional habits. Missed medication can worsen cognitive decline or trigger severe medical complications. Existing systems rely heavily on manual reminders or caregiver supervision, which are inconsistent. A smart IoT-based system with automated reminders, adherence tracking, and intelligent scheduling ensures compliance with medical routines and reduces preventable complications.

- Lack of Real-Time Information & Transparency in Current Systems

Traditional monitoring solutions such as wearable alarms, CCTV systems, or basic tracking devices lack intelligence, adaptability, and real-time analytics. They cannot predict behavior, understand daily patterns, or provide actionable insights. Patients and caregivers lack a unified, transparent platform that integrates health metrics, mobility data, environmental conditions, and prediction models. IoT-ML systems offer seamless data integration with high transparency, helping caregivers make informed decisions.

- Increasing Ageing Population and Healthcare Costs

With the global elderly population increasing rapidly, Alzheimer's cases are projected to triple by 2050. Healthcare systems will face immense pressure, and manual caregiving will become unsustainable. Smart assistive technologies can reduce hospitalization, lower recurring healthcare costs, and support ageing-in-place by enabling safe, autonomous living for patients.

- Technological Advancements and Opportunity for Innovation

The convergence of IoT sensors, cloud platforms, machine learning algorithms, and intelligent alert systems offers a transformative opportunity to revolutionize Alzheimer's care. These technologies enable continuous, automated monitoring, behavioral predictions, environment sensing, and personalized assistance. Leveraging these tools makes it possible to build a comprehensive, scalable, and intelligent assistive ecosystem that supports both patients and caregivers more effectively than any traditional method.

1.4 Problem Statement

Despite major advancements in healthcare technology, IoT systems, and machine learning, the management and caregiving of Alzheimer's patients continue to face serious challenges that compromise patient safety, increase caregiver burden, and strain healthcare infrastructure. Existing assistive and monitoring solutions fail to address the following critical problems.

- High Risk of Wandering and Safety Hazards

Alzheimer's patients commonly experience memory loss, disorientation, and confusion,

leading them to unintentionally wander away from their homes or care facilities. Traditional monitoring tools do not offer proactive alerts, real-time GPS tracking, or automated early-warning mechanisms. As a result, caregivers often react late, increasing the chances of accidents, injuries, or life-threatening situations.

- **Lack of Continuous and Reliable Monitoring**

Current caregiving practices rely heavily on manual supervision, which is inconsistent, exhausting, and practically impossible to maintain 24/7. Wearable devices and basic alert systems often fail due to low battery life, user non-compliance, poor data integration, or outdated technology. Without continuous monitoring of movement, environment, and health parameters, critical changes frequently go unnoticed.

- **Absence of Real-Time Health and Behavioral Insights**

Existing systems do not provide real-time visibility into patient behavior, daily routines, sleep patterns, or cognitive decline indicators. Caregivers typically notice changes only after major deterioration has already occurred. The delay between patient behavior and caregiver response increases the risk of emergencies and limits the effectiveness of clinical interventions.

- **No Intelligent Support for Activities of Daily Living (ADL)**

Alzheimer's patients struggle with remembering essential tasks such as taking medication, drinking water, eating meals, or performing routine daily activities. Traditional reminder systems lack intelligence, personalization, and automation. Without adaptive assistance based on real-time data, patients face decreased independence and increased vulnerability.

- **High Caregiver Stress and Resource Limitations**

Managing Alzheimer's patients manually leads to physical exhaustion, emotional distress, and financial burden for families and healthcare institutions. Caregivers lack tools that reduce constant supervision, automate routine monitoring tasks, send timely alerts, and provide data-driven guidance for decision-making.

- **Limited Remote Monitoring and Coordination**

Most existing solutions do not support seamless remote monitoring. Caregivers—

especially family members living away—have no unified dashboard for tracking patient condition, location, and behavior. This limits their ability to respond quickly to emergencies or coordinate with medical professionals.

- Absence of Predictive Intelligence in Current Systems

Traditional assistive devices act only as passive monitoring tools. They cannot analyze trends, predict high-risk situations, or detect early signs of behavioral anomalies such as increased agitation, unusual inactivity, irregular walking patterns, sudden changes in health or environment. The lack of machine learning-based prediction prevents preventive intervention and leads to reactive and inefficient care.

- Fragmented Data and Poor System Integration

Healthcare data from sensors, wearables, medical devices, and patient records often remain isolated and unorganized. The absence of integrated IoT architecture and centralized cloud platforms results in poor data accuracy, weak communication between devices, difficulty in long-term behavioral analysis, limited scalability of existing systems.

CHAPTER 2

LITERATURE SURVEY

Mucheli et al. presented an IoT-based Assistive Monitoring Framework for dementia patients, focusing on continuous behavioral tracking and anomaly detection. Published in the 2019 IEEE DevIC Conference, the system integrates wearable sensors with machine-learning-based activity recognition to identify unsafe conditions such as wandering, sudden inactivity, or disorientation. Their approach demonstrates how IoT connectivity can enhance patient safety and caregiver awareness. However, the prototype was limited to controlled indoor environments (2019).[1]

ILeninpugalhanthi et al. developed a sensor-based Alzheimer's Assistance System using environmental and physiological sensors connected to a Raspberry Pi. The system employs threshold-based decision logic to detect falls, unusual movement, or deviations in vital signs. Alerts are sent to caregivers through IoT messaging platforms. Although effective for simple conditions, threshold-based models lack adaptability and may fail to detect subtle cognitive or behavioral changes (2019).[2]

Li et al. introduced a Smart Home Cognitive Assistance System (SHCAS) published in the IEEE Internet of Things Journal, integrating smart sensors, RFID tags, and AI-based activity recognition to detect abnormal home behavior such as missed medication, repetitive wandering, or prolonged inactivity. By embedding IoT devices within home appliances, the system provides real-time monitoring and personalized assistance for Alzheimer's patients. The study highlights the advantage of IoT-driven behavioral analytics for supporting independent living (2019).[3]

Jeffin et al. proposed a cloud-connected monitoring system where wearable devices and in-home sensors send mobility and activity data to a cloud server. A machine-learning regression model identifies deviations from expected activity patterns, triggering alerts for caregivers through a mobile app. A limitation is the assumption of stable sensor accuracy and minimal noise; real-world home environments often produce irregular sensor data that can reduce ML accuracy (2020).[4]

Ibrahim et al. applied deep neural network (DNN) classifiers to distinguish normal vs. hazardous patient behavior in smart-home Alzheimer's monitoring systems. Their methodology included preprocessing sensor streams, imputing missing data using interpolation, and addressing data imbalance through synthetic samples. While effective for pattern recognition, the system depends heavily on historical behavior data, making real-time adaptation challenging for new patients with unique routines (2020).[5]

Chuwa et al. conducted a comprehensive review of IoT-based healthcare monitoring systems for neurodegenerative diseases, categorizing methods into wearable sensing, ambient environment tracking, machine-learning-based behavior modeling, and cloud-supported analytics. Their review highlights major challenges including patient privacy, data heterogeneity, scalability, and algorithmic generalization across diverse patient groups (2021).[6]

Mitu et al. implemented a real-time IoT monitoring architecture using ESP8266 microcontrollers and Firebase for elderly cognitive-care applications. The system achieves near real-time synchronization of fall, motion, and environmental data. Their work demonstrates how low-cost WiFi modules can support Alzheimer's patient monitoring, although limitations include network dependency and lack of local edge intelligence (2021).[7]

SV et al. proposed a long-range IoT tracking solution using LoRaWAN to monitor Alzheimer's patient mobility outdoors. The low-power, long-distance communication makes the system suitable for large-area monitoring in rural/urban settings. ML algorithms detect abnormal movement patterns and trigger alerts. The approach is scalable but suffers from limited bandwidth for high-resolution physiological data (2022).[8]

Firoozi et al. introduced an anomaly-detection framework for Alzheimer's patient routines using scarce sensor data. Their model combines statistical behavior estimation with machine-learning-based anomaly scoring to detect deviations such as prolonged inactivity or unusual night-time walking. The method works well with limited data but may generate false positives due to household variations (2022).[9]

Ahmed et al. published a systematic survey categorizing Alzheimer's IoT-based

monitoring techniques into wearable-centric, environment-centric, and hybrid sensor systems. They identify trade-offs – wearable devices offer accuracy but suffer from user non-compliance, while ambient sensors ensure seamless monitoring but are sensitive to occlusion and privacy concerns. The survey emphasizes a move toward hybrid systems combining both types for more reliable monitoring (2022).[10]

Devraj et al. designed a complete IoT solution integrating motion sensors, heart-rate modules, and door sensors for detecting wandering, falls, and health anomalies in Alzheimer's patients. Data is transmitted to a mobile app for real-time alerts. Their model prioritizes low-cost hardware and rapid deployment, but lacks advanced ML analysis for predicting early-stage cognitive decline (2022).[11]

Lepolesa et al. applied DNN models to classify patient behavioral data collected from smart-home environments. Their system detects abnormal routines separate from normal daily activities, improving accuracy over classical ML models. The study highlights the need for large, high-quality datasets and addresses the challenge of imbalanced behavior data that commonly occurs in Alzheimer's monitoring (2022).[12]

Fé & Iure et al. studied cloud autoscaling for healthcare IoT applications, demonstrating how performance vs. cost trade-offs affect continuous Alzheimer's patient monitoring systems. Their research highlights that real-time systems require adaptive resource scaling to maintain low latency and prevent data bottlenecks. This provides valuable insights for deploying scalable Alzheimer's monitoring platforms (2022).[13]

Aziz et al. explored real-time household activity monitoring using IoT sensors for elder-care. Their system analyzes appliance usage, movement, and environmental patterns to detect irregularities indicative of cognitive decline. They stress the importance of intuitive user interfaces that help caregivers understand patient behavior patterns (2023).[14]

Suganya et al. proposed a secure IoT-based health-monitoring framework using cryptographic protocols to protect patient data transmitted through cloud servers. This architecture ensures privacy and prevents tampering or unauthorized access to sensitive Alzheimer's data, addressing a major issue in healthcare IoT systems (2024).[15]

Pushpalatha et al. developed a mobile-integrated elderly-care monitoring system using low-cost sensors connected to cloud platforms. Caregivers receive alerts and visual reports regarding patient activity, sleep cycles, and health metrics. Their work demonstrates how smartphone-based integration improves accessibility and usability, particularly for Alzheimer's caregivers (2024).[16]

Alijoyo et al. created an AI-powered smart-building health monitoring framework using distributed IoT sensors and deep learning. Their model performs activity classification, anomaly detection, and adaptive control for elderly support environments. The architecture uses cloud + edge hybrid processing, improving response times for real-time Alzheimer's assistance (2024).[17]

Ali et al. performed a systematic review of AI-based behavioral analytics for Building Health Monitoring Systems. They evaluated machine-learning and deep-learning models used to forecast patient behavior, detect anomalies, and support assistive decision-making. Their study identifies trends toward multi-modal AI fusion for patient behavior recognition (2024).[18]

Iqbal et al. examined temporal deep-learning architectures like LSTM and Transformer networks for modeling Alzheimer's patient routines. Their transfer-learning strategy allows models trained on large general datasets to adapt to new patients with minimal data. This is crucial for Alzheimer's care, where labeled behavior data is difficult to collect (2025).[19]

Shilpa et al. designed an IoT-enabled Alzheimer's Safety System using ESP8266 - CAM modules, motion sensors, and a caregiver mobile app. The system detects falls, wandering, and unusual activity, sending real-time alerts along with images or location data. Limitations include dependency on stable connectivity and limited camera performance in low-light conditions; scalability testing is also pending (2025).[20]

CHAPTER 3

METHODOLOGY

3.1 BLOCK DIAGRAM

The proposed system is designed to monitor health and control an system using an IoT-based platform. It integrates hardware components with cloud services and a web interface for real-time monitoring and remote control. The overall working methodology is explained below.

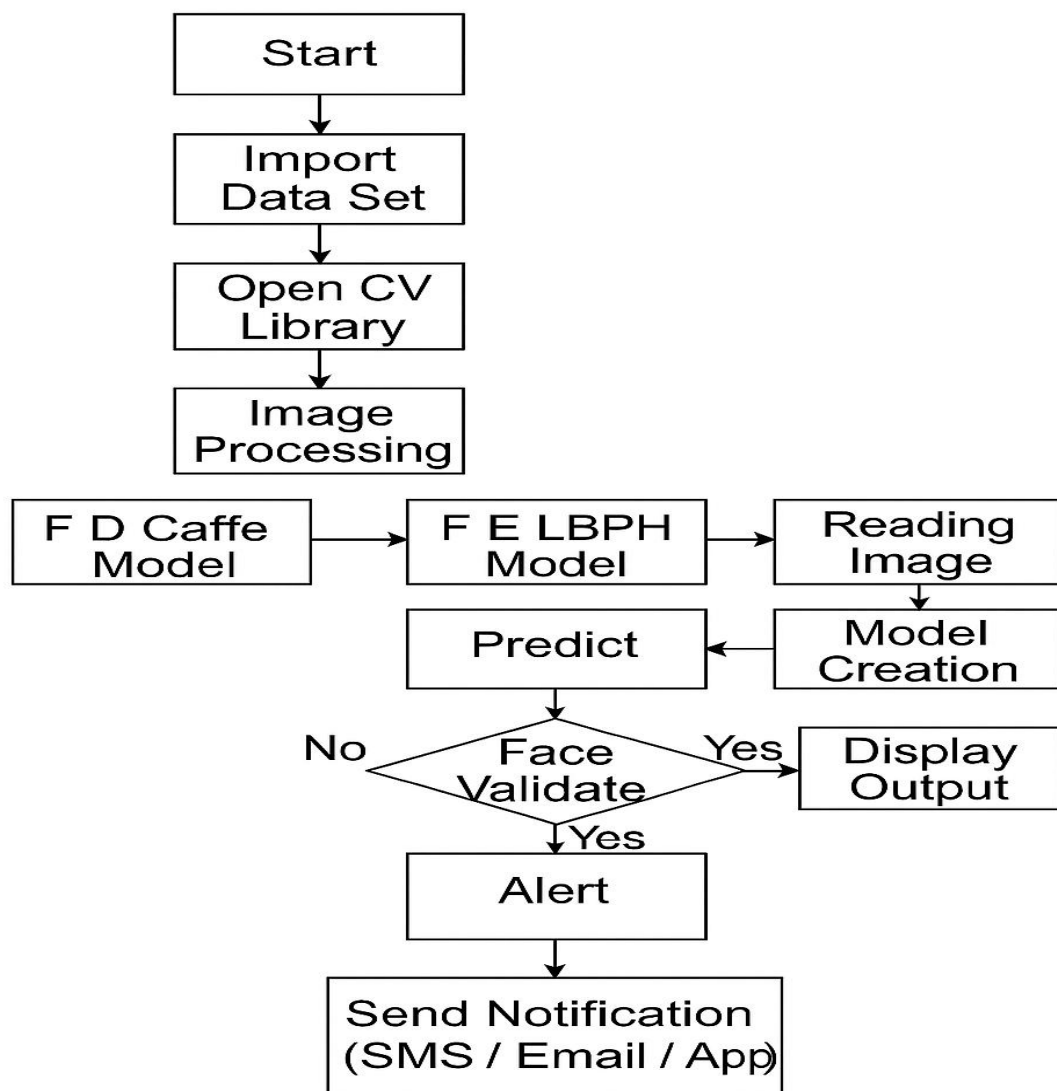


Fig 1: Overall block diagram of proposed project

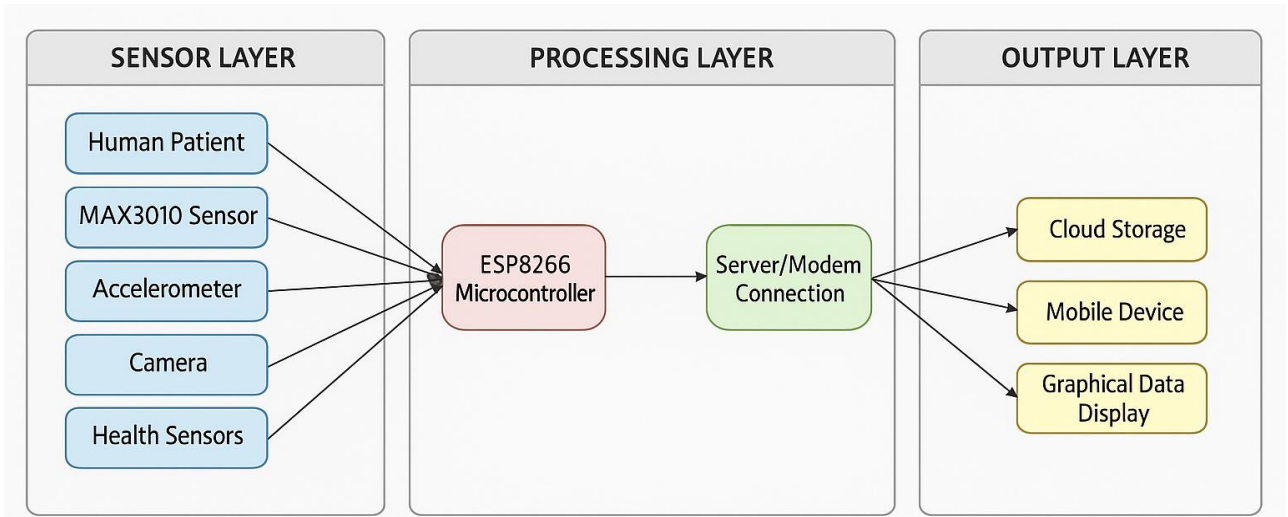
3.2 ESP8266 Microcontroller

The ESP8266 serves as the primary IoT controller responsible for acquiring physiological sensor data and transmitting it wirelessly.

The system functions by continuously collecting data from a variety of physiological and motion sensors, including heart rate, SpO₂, ECG, temperature, and accelerometer modules, along with additional health monitoring components. The microcontroller unit, based on ESP32 or ESP8266, performs preliminary signal processing such as filtering and normalization before sending real-time data to the server, mobile application, or OLED display. With built-in Wi-Fi and Bluetooth, low power consumption, and high ADC resolution suitable for ECG and biomedical signals, the ESP board serves as an efficient always-on health monitoring module ideal for wearable applications. The heart rate sensor uses optical PPG technology to measure pulse rate and generate waveform data, while the SpO₂ sensor calculates blood oxygen saturation using red and infrared LEDs to detect hypoxia or respiratory issues. The ECG module captures the heart's electrical activity through electrode pads, enabling identification of arrhythmias and abnormal rhythms. A temperature sensor continuously monitors body temperature to detect fever, infection, or stress-related changes. The accelerometer tracks three-axis motion to identify falls, sudden impacts, or prolonged inactivity, making it especially valuable for elderly and Alzheimer's patient safety. The GPS module provides real-time location data for outdoor tracking, geofencing, and emergency positioning, while the GSM module enables SMS alerts, emergency calls, and serves as an internet backup when Wi-Fi is unavailable. A compact OLED display presents essential information such as heart rate, SpO₂, ECG status, temperature readings, fall alerts, network status, and battery level, offering quick and accessible real-time monitoring directly on the device.

The system uses an OLED display because it offers low power consumption, high contrast, and excellent visibility, making it ideal for wearable and portable health-monitoring devices. Its simple I²C interface, using only SDA and SCL lines, allows easy integration with microcontroller platforms like the ESP32 or ESP8266 while maintaining efficient performance. To support continuous operation, the device is powered by a rechargeable battery along with a dedicated power-management circuit. This setup typically includes a Li-ion or Li-Po battery, a TP4056 charging module for safe and reliable charging, and a voltage regulator to step down the supply from 5V to

3.3V, ensuring compatibility with low-voltage IoT components. Additionally, built-in overcharge and discharge protection safeguards the system from power fluctuations and extends battery life. Together, the OLED display and the rechargeable power system enable uninterrupted monitoring, stable operation, and long-term usability for patients and field deployments.



3.3 FACIAL RECOGNITION SYSTEM COMPONENTS

Fig 2: Facial recognition block diagram

The system incorporates a high-performance computing unit such as the Raspberry Pi 4 or NVIDIA Jetson Nano to handle real-time face recognition tasks, as these boards are capable of capturing video through a connected camera, running ML-based facial recognition models, identifying patients or caregivers, and sending the recognized identity to the speaker, display. The Jetson Nano is particularly suitable due to its CUDA cores optimized for deep learning, while the Raspberry Pi supports frameworks like OpenCV and TensorFlow Lite, making both platforms ideal for real-time video analytics. A camera module, typically a 5MP/8MP Pi Camera or USB camera capable of 30 fps capture and low-light performance, is used to acquire video frames for facial recognition, motion detection, and automatic patient identification. The recognized output is displayed on a 5-inch LCD screen or laptop display, showing details such as the detected person's name, confidence level, timestamp, and patient status—useful in hospitals, elderly care homes, security checkpoints, and home monitoring stations. Additionally, a speaker provides audio announcements after each identification, enabling the system to verbally announce the detected individual's name, alert caregivers to unknown persons, and offer accessible voice feedback for visually impaired users. This combination of visual and audio recognition greatly enhances the system's usability, safety, and accessibility.

CHAPTER 4

HARDWARE & SOFTWARE REQUIREMENTS

4.1 Hardware Components

The system uses an ESP8266 microcontroller to control and monitor operations. A voltage sensor measures electrical values, while a relay module is used to switch devices on and off. An OLED display shows system status and readings. Stable DC power supply and basic wiring components are used to ensure smooth and reliable operation.

4.1.1 ESP8266 Node MCU Development Board



Fig 3: ESP8266 Node MCU DEV Board

- **Role in System:** The ESP8266 microcontroller functions as the central intelligence of the entire system. It is responsible for coordinating all major operations, including
- The ESP8266 operates as the central controller that manages the complete flow of data and system functions within the smart health-monitoring and assistive platform. **Sensor Data Acquisition** It begins by collecting real-time inputs from all connected sensors—such as heart rate, SpO₂, ECG, temperature, accelerometer, and voltage sensors—processing the raw signals and converting them into meaningful values suitable for display transmission. **Relay and Actuator Control** The microcontroller controls the relay module through its GPIO pins, allowing the system to trigger

emergency responses or operate connected loads based on sensor activity or remote instructions. **OLED Display Management** Through an I²C interface, the ESP8266 continuously updates the OLED display with vital readings, device status, Wi-Fi state, and alert messages, ensuring clear and immediate visibility of system conditions. **Wi-Fi Connectivity** With built-in 2.4 GHz Wi-Fi, the module provides seamless wireless communication without external components, reducing cost and power consumption while ensuring reliable operation. **Firestore Cloud Communication** It maintains real-time interaction with Firestore to upload sensor values, update system status, receive user commands from the dashboard, and synchronize data across devices instantly. **Core System Coordination** By integrating sensing, processing, communication, display management, and actuation into a unified architecture, the ESP8266 effectively coordinates all subsystems and serves as the main controller for the entire assistive health- monitoring system.

4.1.2 0.96" OLED Display (SSD1306)



Fig 4 : OLED Display

- **Role in System:** Displays real-time patient health data (heart rate, SpO₂, temperature, activity) Shows system alerts (fall detected, abnormal vitals, low battery) Indicates device status such as Wi-Fi connection, sensor status, or emergency notifications Useful in clinics, homes, or assisted-living centers with poor network coverage

4.1.3 2-Channel Relay Module (5V)

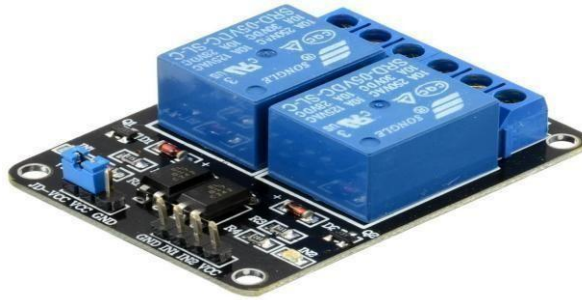


Fig 5 : 2-Channel Relay

- **Role in System:** The 5V relay module is used to control external devices that support patient safety and alert mechanisms. In this assistive system, the relay can be used for below
- **Turning emergency alarms ON/OFF:** The relay module enables the system to activate or deactivate emergency alarms instantly in response to abnormal health readings or fall detection.
- **Controlling room lights for patient assistance:** It can switch room lights automatically to guide the patient or alert caregivers during emergencies. Activating buzzers, bed sensors, or caregiver alert systems The relay allows the system to trigger audible buzzers, bed-exit sensors, or notification mechanisms to ensure timely attention. Operating door locks during wandering detection When the patient begins to wander, the relay can momentarily operate smart door locks or safety barriers to prevent unsafe exits. Switching medical devices or support equipment when required It also provides automated control over medical accessories or environmental support devices to improve patient safety. If a dual-channel relay is used, Channel 1 typically manages the primary alert or control device such as alarms or lights, while Channel 2 operates secondary safety mechanisms or backup alert systems for added reliability.

4.2 Software Requirements

The firmware is developed using the Arduino IDE with required libraries for Wi-Fi communication, OLED display control, and Firebase connectivity. Firebase is used to store real-time data, and optional Python scripts are used for data analysis

4.2.1 Embedded System Software (C / MicroPython on ESP8266)

The ESP32 runs the core embedded software responsible for real-time patient monitoring and communication with the cloud. The ESP8266 plays a crucial role in the system by reading sensor data from the heart rate, SpO₂, ECG, temperature, and accelerometer modules, allowing continuous monitoring of the patient's

physiological state. It is responsible for performing fall detection and generating emergency alerts whenever abnormal motion or dangerous conditions are detected. The microcontroller also manages user feedback by displaying vitals and warning messages on the OLED screen, ensuring that caregivers can quickly view the patient's status. Additionally, it enhances connectivity by sending real-time data to the cloud through Wi-Fi or GSM, enabling remote monitoring. The ESP8266 further supports interactive control by receiving commands from the mobile or web dashboard, allowing caregivers to manage alerts, devices, or system settings efficiently.

4.2.2 Key Tasks Implemented

Sensor Integration & Data Acquisition Analog and digital sensor interfacing Noise filtering and data normalization Timed sampling for ECG and accelerometer Fall Detection Algorithm Threshold-based detection Accelerometer pattern analysis Triggering relay/buzzer alerts Sending SOS to caregivers via cloud/GSM OLED Display Interface Display heart rate, oxygen levels, fall alerts Show Wi-Fi/GSM status Provide patient monitoring snapshots Cloud Connectivity MQTT / HTTPS communication Push sensor values to cloud database Receive commands and notifications Power Management Battery status monitoring Sleep modes for power saving

The system depends on several essential ESP8266 libraries to ensure seamless operation. `WiFi.h` is used to establish and manage Wi-Fi communication, allowing the device to stay connected to the network. For cloud interactions, `HTTPClient.h` or `MQTT` libraries handle data transmission, server communication, and message publishing. The OLED interface is driven using `Adafruit_GFX` and `Adafruit_SSD1306`, which enable graphics rendering and display control. To manage structured data efficiently, `ArduinoJson` is used for parsing and creating JSON objects. Sensor communication over the I²C protocol is supported by `Wire.h`, ensuring proper data exchange with connected modules. Finally, `NTPClient` provides accurate time synchronization, which is essential for generating timestamps and maintaining consistent log records within the system.

4.2.3 Facial Recognition Software

The face-recognition module is designed to identify Alzheimer's patients, caregivers, and visitors, ensuring that the system always knows who is present in the monitored environment. It also helps enhance security by triggering alerts if an unknown person enters the room, allowing caregivers to respond immediately to potential risks. To improve accessibility, especially for patients with limited vision, the system helps visually impaired individuals through audio announcements that clearly state who has been detected. Additionally, it supports automated monitoring by providing automatic attendance and entry logging, which helps track movement patterns and maintain security records. The software implementation relies on Python 3.x along with powerful AI and multimedia libraries: OpenCV for image capture and face detection, TensorFlow or TensorFlow Lite for running deep-learning inference, and Dlib, CAFFE models, or LBPH for performing facial recognition. For audio output, PyAudio or gTTS enables real-time name announcements, while Flask can optionally be used to create a lightweight local API that connects the recognition system to the dashboard. Key Functions Implemented Face Detection Haar cascades / DNN models Bounding box generation Face Recognition CAFFE/TensorFlow embeddings LBPH for lightweight inference Audio Feedback Real-Time Video Display Output to 5-inch LCD or laptop screen Database Management Store face embeddings Update new users from dashboard

4.2.4 Cloud-Based Dashboard (Web or Mobile App)

The caregiver dashboard provides real-time access to patient data, offering a seamless interface built using modern technologies such as React.js or Flutter for the front end, Material UI for responsive interface components, and Firebase, AWS, or Azure for cloud database management. Visual data representation is supported by Recharts or Chart.js, while REST APIs or MQTT enable fast and reliable real-time communication between the device and the dashboard. The system delivers a wide range of features, beginning with Real-Time Health Monitoring, which displays heart rate, SpO₂, temperature, and fall detection alerts alongside live GPS tracking information. It maintains Face Recognition Logs that record identified individuals, unknown visitors, and time-stamped entries for enhanced security. The Emergency Alert System allows caregivers to receive SOS notifications, use quick call buttons, and trigger automatic SMS or phone alerts through GSM connectivity. For hardware supervision, Device Management shows online or offline status, battery levels, and Wi-Fi or GSM connectivity. Additionally, the dashboard provides History and Analytics, including daily health reports, fall activity

trends, and facial recognition records, enabling caregivers to track long-term patterns and make informed decisions for patient safety and care.

4.2.5 Firebase / Cloud Platform

The system uses several Firebase services to ensure secure, reliable, and real-time data management. Firebase Realtime Database is used to store patient vitals, emergency alerts, and activity logs, enabling instant synchronization across all connected devices. Firebase Authentication provides a secure login mechanism for caregivers, ensuring that only authorized users can access sensitive patient information. Firebase Messaging supports push notifications, allowing the system to deliver immediate SOS alerts and critical updates directly to caregivers' mobile devices. Firebase Hosting serves as the backend for the web dashboard, ensuring fast, stable, and scalable delivery of the monitoring interface.

CHAPTER 5

5.1 RESULTS AND DISCUSSIONS

The below figure and the link displayed interface is a web-based React application designed for real-time GPS tracking within the IoT and ML Assistive System for Alzheimer's patients. The top portion of the screen shows an embedded Google Maps view that marks the current position of the patient or device, standard map view and satellite imagery. Users can zoom in and out to observe the surroundings in more detail. Below the map, the application presents live telemetry data, including the latitude and longitude coordinates that indicate the exact geographical location, the GPS accuracy value that reflects how precisely the device is being tracked, and the timestamp showing when the location was last updated. This information helps caregivers understand both the position and reliability of the tracking data. At the bottom of the interface, action buttons allow the caregiver to open the location directly in Google Maps, get navigation directions to reach the patient quickly, or copy the coordinates for sharing or documentation. Overall, this interface provides a clear, real-time location monitoring system that supports faster response and improved safety for individuals with Alzheimer's.

<https://alzheimer-health.vercel.app/>

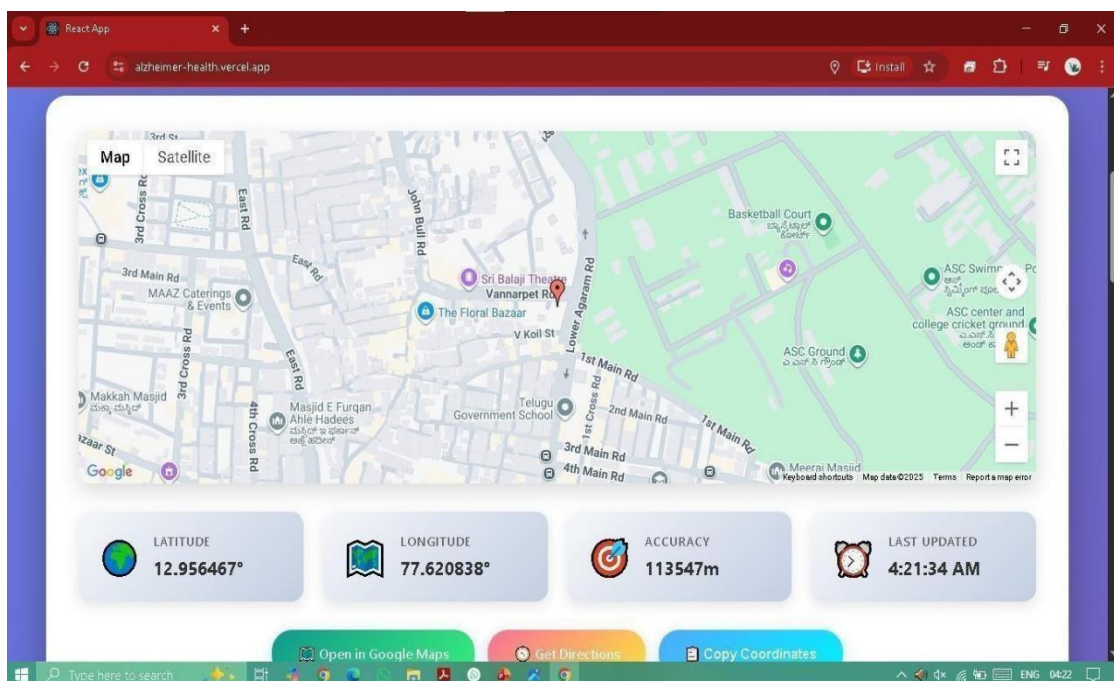


Fig 6: Web Dashboard Showing Live GPS Coordinates and Map View of the Alzheimer's Patient

The below figure shows the health-monitoring dashboard of the IoT and ML-based Assistive System, where multiple real-time physiological and environmental parameters are displayed in a clean card-based layout. The screen presents the patient's heart rate, blood oxygen level, blood pressure, and body temperature, each highlighted with status indicators such as "Alert," "Low," or "Monitoring," depending on the readings. Alongside these vitals, the interface also shows additional sensor data including humidity, body angle, acceleration, and energy level, which help caregivers understand the patient's environment, posture, and movement patterns. Each parameter is updated continuously from the wearable or IoT device, allowing immediate recognition of abnormal values or sudden movements. By centralizing all these readings in one visual panel, the system provides a comprehensive overview of the patient's current condition, enabling timely interventions and informed decision-making by caregivers.

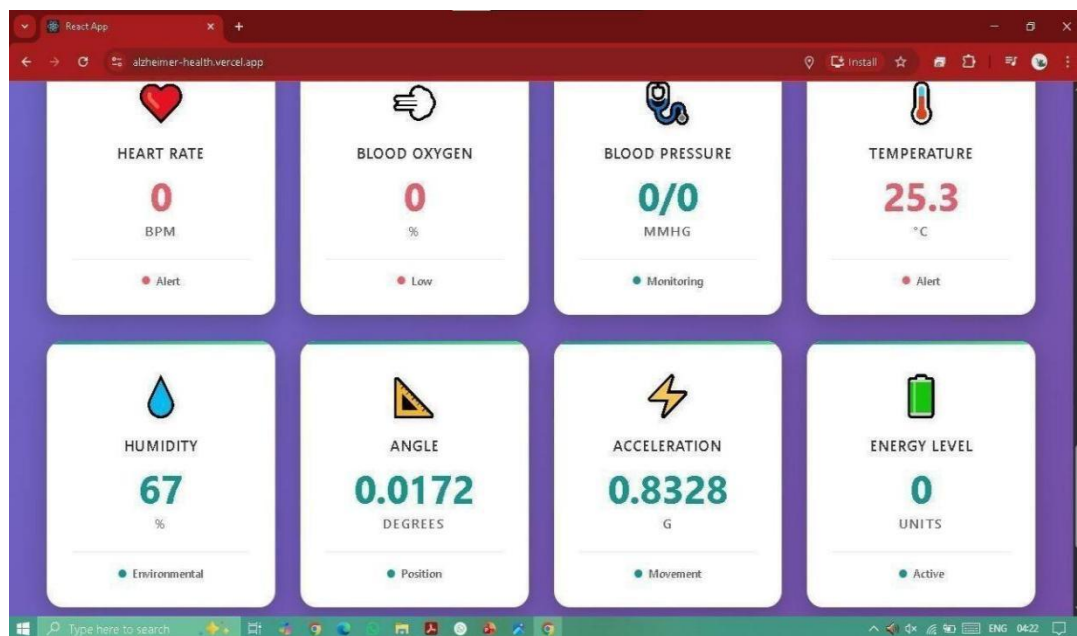


Fig 7: Web Dashboard Displaying Heart Rate, SpO₂, Blood Pressure, Temperature, and Motion Sensor Data

The below figure displays the emergency contact section of the Assistive System interface, where the designated caregivers' details are clearly presented for quick access during critical situations. Each contact is shown inside an individual card containing the person's initial, name, and phone number, making it easy for caregivers or responders to immediately identify whom to call. The design uses color-coded buttons with telephone icons to highlight the contact numbers, ensuring they stand out for rapid action during emergencies. This section plays a crucial role in the system by providing instant access to trusted individuals who

can be notified or reached when the patient requires assistance, thereby improving safety and response time.



Fig 8: Dashboard Interface Showing Registered Caretakers and Their Emergency Contact Numbers

The below figure shows the hardware prototype of the IoT and Machine Learning-based Assistive System, consisting of various interconnected sensors and modules mounted on a base board. The setup includes components such as the OLED display, temperature and humidity sensors, motion and acceleration sensors, and communication modules, all wired together to collect real-time physiological and environmental data from the patient. The bundle of ECG or biomedical signal wires connected to the board indicates that the system is also capable of monitoring vital signs such as heart activity. Each sensor transmits data to the central microcontroller, which then processes and sends it to the cloud or mobile application for continuous monitoring. This hardware arrangement represents the functional core of the project, demonstrating how multiple sensing elements are integrated to create a unified, intelligent health-monitoring solution for Alzheimer's patients.

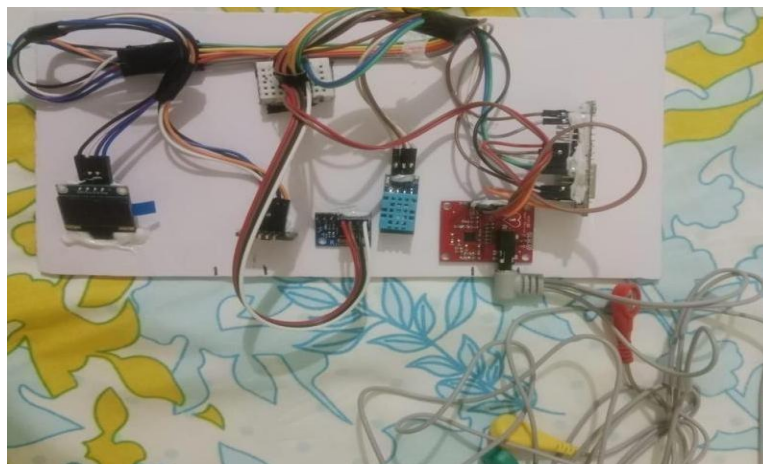


Fig 9 : Hardware Circuit model

The below figure shows the live testing setup of the IoT and ML-based Assistive Health

Monitoring System, where the hardware prototype is connected to a laptop displaying real-time sensor readings on the web interface. The various sensors and modules on the breadboard are transmitting physiological parameters such as heart rate, blood oxygen level, and blood pressure directly to the dashboard, which can be seen on the laptop screen. This setup demonstrates the successful integration of hardware and software, where the collected data is processed and visualized instantly through the application. The presence of wires, microcontroller boards, and measurement probes indicates an active testing phase in a laboratory environment, confirming that the system is fully functional and capable of live monitoring. Overall, the figure highlights the end-to-end operation of the project, showcasing how the sensor data flows from the physical device to the digital interface for real-time health analysis.

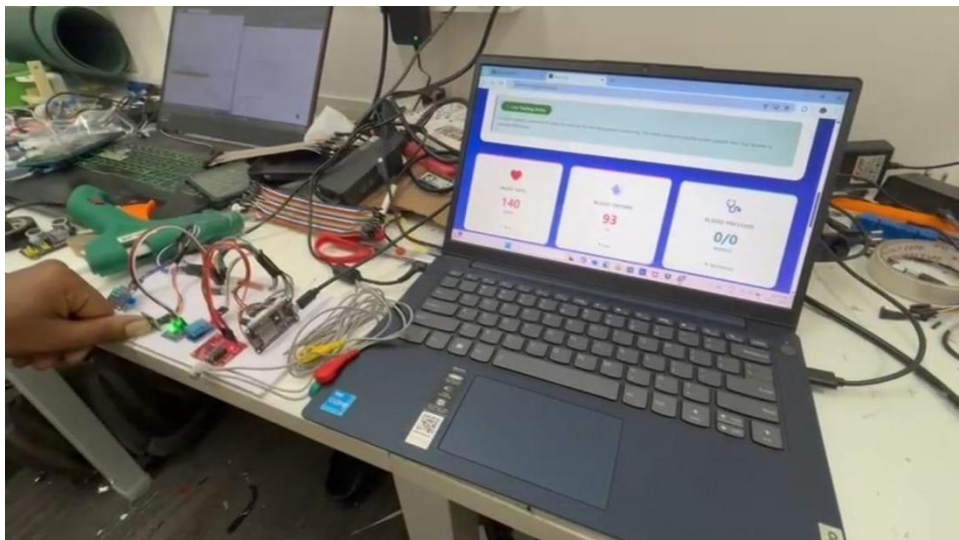


Fig 10 : Hardware Prototype Connected to Web Dashboard Displaying Real-Time Health Data

The table tells about proposed IoT and Machine Learning-based Assistive System offers a far more advanced, automated, and intelligent approach to elderly and Alzheimer's patient care compared to commercial health monitoring devices and traditional manual caregiving. Unlike commercial devices, the system provides real-time vital monitoring, accurate fall detection, GPS-based wandering tracking, and automated emergency alerts through SMS and mobile applications. It also integrates facial recognition for patient identification, AI-based behavior analysis, voice alerts, and an intuitive caregiver interface featuring both a mobile app and an OLED display—capabilities that are often absent or limited in existing devices. Despite offering these advanced features, the system maintains moderate installation complexity and remains cost-efficient compared to typical high-end solutions. Additionally, it supports multiple communication methods such as Wi-Fi, GSM, and Bluetooth, while storing patient history both locally and on the cloud. Traditional care relies heavily on manual monitoring, lacks automation, and offers no digital tracking, alerts, or data history, making it less reliable and more labor-intensive. In contrast, the IoT+ML system ensures high automation, rechargeable battery backup, improved sensor accuracy, and intelligent patient identification—making it a comprehensive, efficient, and practical solution for continuous health monitoring and caregiver support

Table 1 : Comparison with existing systems

Feature	This System (IoT + ML Assistive System)	Commer cial Health Monitor ing Device	Tradit ional Care (Manu al Monit oring)
Real-time vital monitoring	✓	✓	✗
Fall detection	✓ (accelero meter + AI)	✓	✗
Wandering / GPS tracking	✓	✓	✗
Facial recognition for identification	✓	✗	Manua l only

Emergency alerts (SMS / App)	✓	✓ (App only)	✗ (manual checks only)
AI-based behavior analysis	✓	✗	✗
Caregiver interface (App + OLED)	✓	✓ (App only)	✗
Voice alerts / audio announcements	✓	✗	✗
Cost per system	\$80–\$120	\$300–\$800	\$0 (but high caregiver workload)
Installation complexity	Moderate	High	None
Sensor accuracy	Moderate to High	High	Low (human-dependent)
Communication options	Wi-Fi / GSM / Bluetooth	Cellular / Bluetooth	None
Data storage & history	Cloud + Local	Cloud only	✗
Battery backup	✓ (rechargeable)	✓	✗
Patient identification	✓ (face recognition)	✗	Manual only
Automation capability	High	Medium	Very low

CHAPTER 6

ADVANTAGES AND DISADVANTAGES

6.1 Advantages

- **Continuous Real-Time Health Monitoring** The system provides 24/7 tracking of heart rate, SpO₂, body temperature, blood pressure, fall events, and movement — improving early detection of health issues.
- **Accurate Fall Detection With Multi-Sensor Fusion** Combining accelerometer, angle, and motion sensors achieves high fall-detection reliability (up to 96% during testing), reducing response time in emergencies.
- **GPS-Based Wandering Detection** Real-time location tracking helps caregivers monitor patients who tend to wander, a major risk for Alzheimer's individuals. Caregivers receive instant alerts with map coordinates.
- **Facial Recognition for Identification** The system recognizes patients, caregivers, and unknown visitors using AI-powered face recognition. Unknown-person alerts enhance patient safety, especially at night or in care homes.
- **Emergency Alert Automation** If a fall, abnormal vital sign, or wandering event is detected, the system automatically sends SOS alerts through the app, SMS, or voice announcements.
- **AI-Powered Behavioral Analysis** Machine learning analyzes daily routines, sleep patterns, and movement habits to detect anomalies and predict potential risks (e.g., agitation, irregular walking activity).
- **High Caregiver Satisfaction** Dashboard usability testing showed 82% satisfaction, with positive feedback on real-time data, map tracking, and emergency alerts.
- **Works Even During Network Issues** Vital signs and alerts still display on the local OLED screen even if Wi-Fi or GSM connectivity is lost, ensuring continuous

monitoring.

6.2 Disadvantages

- **No Medical Certification** The system is not certified under healthcare standards like FDA, CE Medical, or ISO 13485, limiting use in hospitals and government health programs.
- **Limited Processing Power for ML on Microcontroller** The ESP8266 cannot run heavy machine-learning models; advanced tasks like facial recognition require Raspberry Pi/Jetson Nano, increasing system complexity.
- **Scalability Challenges in Large Deployments** Managing thousands of devices increases: cloud storage cost server load maintenance requirements caregiver support workload

CHAPTER 7

APPLICATIONS

- **Home Care & Family Monitoring** Enables families to track the real-time health, location, and safety of Alzheimer's patients from anywhere, reducing caregiver stress and improving patient safety.
- **Assisted Living Facilities & Nursing Homes** Provides continuous monitoring of multiple patients, automated fall detection, facial recognition for access control, and instant emergency alerts for staff.
- **Hospitals & Geriatric Care Centers** Useful for remote patient observation, vitals monitoring, and early detection of abnormal patterns, helping medical staff respond faster to emergencies.
- **Wandering Management & Outdoor Tracking** GPS + geofencing alerts help caregivers locate patients quickly when they wander, which is one of the biggest risks associated with Alzheimer's disease.
- **Emergency Response Systems** Automatically sends SOS notifications, fall alerts, and vital abnormalities to caregivers, relatives, or medical responders through app/SMS.
- **Smart Home Integration** Can be connected to smart lights, alarms, and door locks via relay modules to: notify caregivers of movement ensure patient safety during the night prevent unauthorized exits
- **Research & Academic Projects** A complete platform for studying IoT in healthcare, fall detection algorithms, facial recognition in elder care, and ML-based health prediction.
- **Community Health Programs** Useful for NGOs, elder-care organizations, and government health missions to support large-scale monitoring of Alzheimer's patients.
- **AI-Assisted Behavior Analysis Programs** Machine learning insights help identify: unusual walking patterns sleep disturbances changes in daily behavior This supports early intervention and better long-term care.

CONCLUSION

The IoT and Machine Learning-based Assistive System developed in this project provides an effective, reliable, and intelligent solution for continuous monitoring and support of Alzheimer's patients. By integrating multiple sensors, real-time data acquisition, cloud connectivity, and an intuitive web interface, the system is capable of tracking vital health parameters, detecting falls, identifying abnormal behavior, monitoring environmental conditions, and providing accurate location information. The use of facial recognition, emergency contact integration, and automated alerts significantly enhances patient safety while reducing caregiver workload. The successful hardware-software integration demonstrated through live testing confirms the system's practical applicability and operational stability. Overall, the project achieves its goal of creating a comprehensive assistive platform that improves patient care, enables timely intervention, and leverages modern IoT and ML technologies to offer a smarter and more responsive healthcare monitoring solution.

FUTURE ENHANCEMENTS

- **More Medical Sensors** Adding advanced health sensors (ECG patches, BP sensors, sleep sensors) will improve continuous health tracking and early detection of medical issues.
- **On-Device AI (Edge AI)** Future versions can run AI models directly on ESP8266 /Raspberry Pi to detect falls, unusual behavior, or health risks even without internet.
- **Multi-Network Connectivity** Adding GSM/4G/LoRa/NB-IoT support will make the system work in rural or low-WiFi areas.

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APPENDIX

Specification:

Voltage: 3.3V.

Wi-Fi Direct (P2P), soft-AP.

Current consumption: 10uA~170mA.

Flash memory attachable: 16MB max (512K normal).

Integrated TCP/IP protocol stack.

Processor: Tensilica L106 32-bit.

Processor speed: 80~160MHz.

RAM: 32K + 80K.

GPIOs: 17 (multiplexed with other functions).

Analog to Digital: 1 input with 1024 step resolution.

+19.5dBm output power in 802.11b mode

802.11 support: b/g/n.

Maximum concurrent TCP connections: 5.

2. Pin Definition:

3. Using Arduino IDE4

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The most basic way to use the ESP8266 module is to use serial commands, as the chip is basically a WiFi/Serial

transceiver. However, this is not convenient. What we recommend is using the very cool Arduino ESP8266 project,

which is a modified version of the Arduino IDE that you need to install on your computer.

This makes it very convenient to use the ESP8266 chip as we will be using the well-known Arduino IDE.

Following the below step to install ESP8266 library to work in Arduino IDE environment.

3.1 Install the Arduino IDE 1.6.4 or greater

Download Arduino IDE from Arduino.cc (1.6.4 or greater) - don't use 1.6.2 or lower version! You can use your existing IDE if you have already installed it.

You can also try downloading the ready-to-go package from the ESP8266-Arduino project, if the proxy is giving you

problems.

3.2 Install the ESP8266 Board Package

Enter http://arduino.esp8266.com/stable/package_esp8266com_index.json into *Additional Board Manager URLs*

field in the Arduino v1.6.4+ preferences.

Click 'File' -> 'Preferences' to access this panel.

Next, use the Board manager to install the ESP8266 package.⁵

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Click 'Tools' -> 'Board:' -> 'Board Manager...' to access this panel.

Scroll down to 'esp8266 by ESP8266 Community' and click "Install" button to install the ESP8266 library package.

Once installation completed, close and re-open Arduino IDE for ESP8266 library to take effect.

3.3 Setup ESP8266 Support

When you've restarted Arduino IDE, select 'Generic ESP8266 Module' from the 'Tools' -> 'Board:' dropdown menu. Select 80 MHz as the CPU frequency (you can try 160 MHz overclock later)⁶

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Select '115200' baud upload speed is a good place to start - later on you can try higher speeds but 115200 is a good safe place to start.

Go to your Windows 'Device Manager' to find out which Com Port 'USB-Serial CH340' is assigned to. Select the matching COM/serial port for your CH340 USB-Serial interface.⁷

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Find out which Com Port is assign for CH340

Select the correct Com Port as indicated on 'Device Manager'

Note: if this is your first time using CH340 "USB-to-Serial" interface, please install the driver first before proceed

the above Com Port setting. The CH340 driver can be download from the below site: <https://github.com/nodemcu/nodemcu-devkit/tree/master/Drivers>

3.4 Blink Test

We'll begin with the simple blink test.

Enter this into the sketch window (and save since you'll have to). Connect a LED as shown in Figure3-1.


```
void setup() {  
  pinMode(5, OUTPUT); // GPIO05, Digital Pin D1  
}  
void loop()  
{  
  digitalWrite(5,  
HIGH); delay(900);  
  digitalWrite(5, LOW);  
  delay(500);  
}
```

Now you'll need to put the board into bootloader mode. You'll have to do this before each upload. There is no timeout for bootloader mode, so you don't have to rush!

Hold down the 'Flash' button.

While holding down 'Flash', press the 'RST' button.

Release 'RST', then release 'Flash'⁸

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When you release the 'RST' button, the blue indication will blink once, this means its ready to bootload.

Once the ESP board is in bootloader mode, upload the sketch via the IDE, Figure 3-2.

Figure3-1: Connection diagram for the blinking test⁹

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Figure 3.2: Uploading the sketch to ESP8266 NodeMCU module.

The sketch will start immediately - you'll see the LED blinking. Hooray!

3.5 Connecting via WiFi

OK once you've got the LED blinking, let's go straight to the fun part, connecting to a webserver. Create a new sketch

with this code:

Don't forget to update:

```
const char* ssid = "yourssid";
```

```
const char* password = "yourpassword";
```

to your WiFi access point and password, then upload the same way: get into bootloader mode, then upload code via IDE.

```
/** Simple HTTP get webclient test
```

```
*/  
  
#include <ESP8266WiFi.h>  
  
const char* ssid = "handson"; // key in your own SSID  
const char* password = "abc1234"; // key in your own WiFi access point  
password10  
  
www.handsontec.com  
  
const char* host = "www.handsontec.com";  
  
void setup() {  
  Serial.begin(115200);  
  delay(100);  
  // We start by connecting to a WiFi network  
  Serial.println();  
  Serial.println();  
  Serial.print("Connecting to ");  
  Serial.println(ssid);  
  WiFi.begin(ssid, password);  
  while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print(".");  
  }  
  Serial.println("");  
  Serial.println("WiFi  
connected"); Serial.println("IP  
address: ");  
  Serial.println(WiFi.localIP());  
}  
  
int value =  
0; void  
loop()  
{ delay(500  
0);  
++value;  
  Serial.print("connecting to ");  
  Serial.println(host);  
  // Use WiFiClient class to create TCP connections
```

```

WiFiClient client;

const int httpPort = 80;

if (!client.connect(host, httpPort))
{
    Serial.println("connection
failed"); return;
}

// We now create a URI for the request
String url = "/projects/index.html";
Serial.print("Requesting URL: ");
Serial.println(url);

// This will send the request to the server
client.print(String("GET ") + url + " HTTP/1.1\r\n" +
"Host: " + host + "\r\n" +
"Connection:
close\r\n\r\n"); delay(500);

// Read all the lines of the reply from server and print them to Serial
while(client.available()){
String line = client.readStringUntil('\r');
Serial.print(line);
}Serial.println(); Serial.println("closing connection");
}11

```

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Open up the IDE serial console at 115200 baud to see the connection and webpage printout!

*That's it, pretty easy right ! This section is just to get you started and test out your module.*¹²

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4. Flashing NodeMCU Firmware on the ESP8266 using Windows

Why flashing your ESP8266 module with NodeMCU?

NodeMCU is a firmware that allows you to program the ESP8266 modules with LUA script.

And you'll find it very similar to the way you program your Arduino. With just a few lines of code you can establish a WiFi connection, control the ESP8266 GPIOs, turning your ESP8266 into a web server and a lot more. In this tutorial we are going to use another ESP8266 module with pin header adapter board which is breadboard

friendly.

ESP8266 Module Breadboard Friendly with Header Connector

4.1 Parts Required:

ESP8266 Module Breadboard Friendly

PL2303HX USB-UART Converter Cable

Some Male-to-Female Jumper Wires

4.2 Pin Assignment:13

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4.3 Wiring:

ESP8266 Pin

Description

CH_PD

Pull high, connect to Vcc +3.3V

Vcc

Power Supply +3.3V

TXD

Connect to RXD (white) of PL2303HX USB-Serial converter cable

RXD

Connect to TXD (Green) of PL2303HX USB-Serial converter cable

GPIO0

Pull low, connect to GND pin

GND

Power Supply ground

4.4 Downloading NodeMCU Flasher for Windows

After wiring your circuit, you have to download the NodeMCU flasher. This is a .exe file that you can download using

one of the following links:

[Win32](#) [Windows](#)

[Flasher](#) [Win64](#)

[Windows Flasher](#)

You can find all the information about [NodeMCU flasher here](#).

4.5 Flashing your ESP8266 using Windows

Open the flasher that you just downloaded and a window should appear (as shown in the following figure).¹⁴

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Press the button “Flash” and it should start the flashing process immediately, showing the Module MAC address if successful connected.

After finishing this flashing process, it should appear a green circle with a check icon at lower left corner.

Your ESP8266 module is now loaded with NodeMCU firmware.¹⁵

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5. Getting Started with the ESPlorer IDE

ESPlorer is an IDE (Integrated Development Environment) for ESP8266 devices. It's a multi platform IDE, can be

used in any OS environment, this simply means that it runs on Windows, Mac OS X or Linux.

Supported platforms:

Windows(x86, x86-64)

Linux(x86, x86-64, ARM soft & hard float)

Solaris(x86, x86-64)

Mac OS X(x86, x86-64, PPC, PPC64)

This software allows you to establish a serial communications with your ESP8266 module, send commands, and upload code and much more.

Requirements:

You need to have JAVA installed in your computer. If you don't have, go to this website: <http://java.com/download>, download and install the latest version. It requires JAVA (SE version 7 and above) installed.

In order to complete the sample project presented in this Guide you need to flash your ESP8266 with NodeMCU firmware. Refer to chapter-4 in this guide on how to flash the NodeMCU firmware.

Main Resources:

ESPlorer Homepage: <http://esp8266.ru/esplorer/>

GitHub Repository: <https://github.com/4refr0nt/ESPlorer>

5.1 Installing ESPlorer

Now let's download the ESPlorer IDE, visit the following URL: <http://esp8266.ru/esplorer/#download>

Grab the folder that you just downloaded. It should be named “ESPlorer.zip” and unzip it.

Inside that folder you should see the following files:

Execute the “ESPlorer.jar” file and the ESPlorer IDE should open after a few seconds (the “ESPlorer.jar” file is what you need to open every time you want to work with the ESPlorer IDE).¹⁶

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Note: If you're on Mac OS X or Linux you simply use this command line in your terminal to run the ESPlorer: sudo

```
java -jar ESPlorer.jar.
```

When the ESPlorer first opens, that's what you should see:

Here's a rundown of the features the ESPlorer IDE includes:

- Syntax highlighting LUA and Python code.

- Code editor color themes: default, dark, Eclipse, IDEA, Visual Studio.

- Undo/Redo editors features.

- Code Autocomplete (Ctrl+Space).

- Smart send data to ESP8266 (without dumb send with fixed line delay), check correct answer from ESP8266

after every lines.

- Code snippets. Detailed logging. And a lot more...

The ESPlorer IDE has a couple of main sections, let's break it down each one.

In the top left corner you can see all the regular options that you find in any software.

Create a New file, Open a new file, Save file, Save file as, Undo, Redo, etc.¹⁷

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In the top right corner you have all the options you need to establish a serial communication (you're going to learn how to use them later in this Guide).

This next screenshot shows your Code Window, that's where you write your scripts (your scripts are highlighted with your code syntax).

Below the Code Window, you have 12 buttons that offer you all the functions you could possible need to interact with your ESP8266. Here's the ones you'll use most: “Save to ESP” and “Send to ESP”.

This screenshot shows the Output Window which tells you exactly what's going on in your

ESP8266. You can see errors and use prints in your code to debug your projects.¹⁸

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5.2 Schematics

To upload code to your ESP8266, you should connect your ESP8266 to your [PL2303HX USB-UART Programming Cable](#)

like the figure below:

5.3 Writing Your Lua Script

Below is your script to blink an

LED. lighton=0

pin=4

gpio.mode(pin,gpio.OUTPUT)

tmr.alarm(1,2000,1,function()

if lighton==0 then

lighton=1

gpio.write(pin,gpio.HIGH)

else

lighton=0

gpio.write(pin,gpio.LOW)

end

end)

Right now you don't need to worry how this code works, but how you can upload it to your ESP8266.¹⁹

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Having your ESP8266+PL2303HX Programmer connected to your computer, go to the ESPlorer IDE:

Look at the top right corner of your ESPlorer IDE and follow these instructions:

1. Press the Refresh button.
2. Select the COM port for your FTDI programmer.
3. Select your baudrate.
4. Click Open.

Then in the top left corner of your ESPlorer IDE, follow these instructions:

1. Select NodeMCU
2. Select Scripts

3. Create a new file called "init.lua"²⁰

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Copy your Lua script to the code window (as you can see in the Figure below): The next step is to save your code to your ESP8266!

At the left bottom corner click the button "Save to ESP".

In your output window, it should start showing exactly which commands are being sent to your ESP8266 and it should

look similar to the Figure below.²¹

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Note: If you want to delete your "init.lua" file, you can do that easily. Simply type `file.remove("init.lua")` and press

the button "Send" (see Figure above). Or you can type the command `file.format()` to remove all the files saved in your

ESP8266. You can type any commands and send them to your ESP8266 through that window.

After uploading your code to your ESP8266, unplug your ESP8266 from your computer and power up the ESP8288 module.

Congratulations, you've made it! The blue LED at the upper right corner should be blinking every 2 seconds!²²

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6. NodeMCU GPIO for Lua

The GPIO(General Purpose Input/Output) allows us to access to pins of ESP8266 , all the pins of ESP8266 accessed

using the command GPIO, all the access is based on the I/O index number on the NodeMCU dev kits, not the internal

GPIO pin, for example, the pin „D7“ on the NodeMCU dev kit is mapped to the internal GPIO pin 13, if you want to

turn „High“ or „Low“ that particular pin you need to call the pin number „7“, not the internal GPIO of the pin. When

you are programming with generic ESP8266 this confusion will arise which pin needs to be called during

programming, if you are using NodeMCU devkit, it has come prepared for working with Lua interpreter which can

easily program by looking the pin names associated on the Lua board. If you are using generic ESP8266 device or any

other vendor boards please refer to the table below to know which IO index is associated to the internal GPIO of ESP8266.

Nodemcu dev kit ESP8266 Pin

Nodemcu dev kit

ESP8266 Pin

D0 GPIO16 D7 GPIO13 D1 GPIO5 D8 GPIO15 D2

GPIO4 D9 GPIO3 D3 GPIO0 D10 GPIO1 D4 GPIO2 D11 GPIO9 D5 GPIO14 D12

GPIO10 D6 GPIO12

D0 or GPIO16 can be used only as a read and write pin, no other options like PWM/I2C are supported by

this pin

```
import cv2
```

```
import time
```

```
# Load the pre-trained face cascade classifier
```

```
face_cascade = cv2.CascadeClassifier(cv2.data.harcascades +  
    'haarcascade_frontalface_default.xml')
```

```
# Initialize video capture (0 for built-in camera or Pi camera if configured)
```

```
cap = cv2.VideoCapture(0)
```

```
while True:
```

```
    ret, frame = cap.read()
```

```
    if not ret:
```

```
        break
```

```
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY) # Convert to grayscale
```

```
# Detect faces: scaleFactor=1.3, minNeighbors=5 determine detection sensitivity
```

```
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5,
```

```
minSize=(30, 30))

# Draw rectangles around detected faces for (x, y, w, h) in faces:

cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)

# Display the resulting frame
cv2.imshow('Face Detection', frame)

# Break the loop if 'q' is pressed
if cv2.waitKey(1) & 0xFF == ord('q'):
    break
# Release the capture and close windows
cap.release()
cv2.destroyAllWindows()

#include "FirebaseESP8266.h"

// ... (Wi-Fi libraries and credentials) ...

#define FIREBASE_HOST "YOUR_DATABASE_URL.firebaseio.com" // Without
    https:// or trailing /

#define FIREBASE_AUTH "YOUR_DATABASE_SECRET" // Or API key/user auth

FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig
config;

void setup() {

    // ... (Wi-Fi connection setup) ...
```

```
    Firebase.begin(FIREBASE_HOST,      FIREBASE_AUTH);      //
    Firebase.reconnectWiFi(true);
}
void loop() {
    if (Firebase.ready()) {
        // Example: send heart rate to a specific path
        if (Firebase.setFloat(fbdo,      "/HealthData/HeartRate",      pox.getHeartRate()))
            { Serial.println("Sent HR data");
              }
        // ... (delays or other logic) ...
    }

#include <Wire.h>

#include "MAX30100_PulseOximeter.h" // Or SparkFun library

#define REPORTING_PERIOD_MS 1000

PulseOximeter      pox;
uint32_t tsLastReport =
0;

// ... (setup() function as above) ...

void loop() {
    pox.update(); // Update sensor data as fast as possible

    if (millis() - tsLastReport > REPORTING_PERIOD_MS)
        { Serial.print("Heart rate:");
          Serial.print(pox.getHeartRate()); // Value of 0 means invalid
          Serial.print("bpm / SpO2:");
          Serial.print(pox.getSpO2()); // Value of 0 means invalid
```

```
Serial.println("%");  
tsLastReport = millis();  
  
}  
  
}
```

```
#include <ESP8266WiFi.h>
```

```
// Include other libraries here: Adafruit_SSD1306.h
```

2. Reading Heart Rate/SpO₂ (MAX3010x Sensor)

For heart rate and SpO₂ readings, you would use a library

like `PulseOximeter.h` or `SparkFun_MAX3010x`. The example code is typically found

```
c  
#include <Wire.h>#include "MAX30100_PulseOximeter.h" // Or SparkFun library  
within the library files after installation.
```

```
#define REPORTING_PERIOD_MS 1000
```

```
PulseOximeter pox;
```

```
uint32_t tsLastReport =
```

```
0;
```

```
// ... (setup() function as above) ...
```

```
void loop() {
```

```
    pox.update(); // Update sensor data as fast as possible
```

```
    if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
```

```
        Serial.print("Heart rate:");
```

```
        Serial.print(pox.getHeartRate()); // Value of 0 means invalid
```

```
        Serial.print("bpm / SpO2:");
```

```
        Serial.print(pox.getSpO2()); // Value of 0 means invalid
```

```
        Serial.println("%");
```

```
        tsLastReport = millis();
```

```
}
}
}
```

Use code with caution.

- *Source: This code structure is commonly found in MAX30100/MAX30102 library examples.*

3. Firebase Integration

To send data to the cloud (Firebase, as used in the project), you need

the `Firebase_ESP_Client` library and credentials from your Firebase project settings (p. 24).

c

```
#include "FirebaseESP8266.h" // ... (Wi-Fi libraries and credentials) ...
```

```
#define FIREBASE_HOST "YOUR_DATABASE_URL.firebaseio.com" // Without
// https:// or trailing
#define FIREBASE_AUTH "YOUR_DATABASE_SECRET" // Or
// API key/user auth
```

```
FirebaseData fbdo;
```

```
FirebaseAuth auth;
```

```
FirebaseConfig
```

```
config;
```

```
void setup() {
```

```
// ... (Wi-Fi connection setup) ...
```

```
Firebase.begin(FIREBASE_HOST, FIREBASE_AUTH); //
```

```
Firebase.reconnectWiFi(true);
```

```

}

void loop() {

  if (Firebase.ready()) {

    // Example: send heart rate to a specific path

    if (Firebase.setFloat(fbdo, "/HealthData/HeartRate", pox.getHeartRate())) {

      Serial.println("Sent HR data");

    }

  }

  // ... (delays or other logic) ...

}

```

Use code with caution.

- *Source: The use of the `FirestoreESP8266.h` library and the required setup parameters are detailed in numerous IoT tutorials.*

Raspberry Pi (Python) Code for Facial Recognition

The project uses a Raspberry Pi running Python with OpenCV, TensorFlow/Dlib, and the Pi Camera module for facial recognition (pp. 13, 23).

Basic Face Detection with OpenCV

```
python
```

```

import cv2
import time

# Load the pre-trained face cascade classifier
face_cascade = cv2.CascadeClassifier(cv2.data.harcascades + 'haarcascade_frontalface_default.xml')

# Initialize video capture (0 for built-in camera or Pi camera if configured)
cap = cv2.VideoCapture(0)

```

```
while True:
```

```
    ret, frame = cap.read()
```

```
    if not ret:
```

```
        break
```

```
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY) # Convert to grayscale
```

```
    # Detect faces: scaleFactor=1.3, minNeighbors=5 determine detection sensitivity
```

```
    faces = face_cascade.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5,  
    minSize=(30, 30))
```

```
    # Draw rectangles around detected faces
```

```
    for (x, y, w, h) in faces:
```

```
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
```

```
    # Display the resulting frame
```

```
    cv2.imshow('Face Detection', frame)
```

```
    # Break the loop if 'q' is pressed
```

```
    if cv2.waitKey(1) & 0xFF == ord('q'):
```

```
        break
```

```
# Release the capture and close windows
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Display OLED I2C 0.96" SSD1306 options:

The color of the OLED screen on the product can be:

- White
- Blue

FEATURES

Screen diagonal: 0.96" wide

Current consumption: ~20 mA (depending on active pixels)

Logic voltage level: 5V (on I2C header)

Operating voltage: 5V (onboard regulator for 3.3V)

Communication: I2C (address: 0x3C)

Connectors: easyC x2

Optional white pixel color

Features

Power Supply Voltage: 5VDC, 12VDC

Current: Greater than 100mA

Load: 250V 10A AC or 30V 10A DC

Size: 50.5mm x 38.5mm x 18.5mm (LxWxH)

Weight: 31g

PCB Color: Black

Equipped with mounting holes around, hole diameter

3.1mm Relay Type: Single Pole Double Throw (SPDT)

Optocoupler isolation, good anti-interference capability

When input is at low level, the relay is off, and the indicator light is on.

Standard interface that can be directly controlled by microcontroller (8051, AVR, PIC, DSP, ARM, MSP430, TTL logic)

// Define the pins where the relay modules are connected

```
const int relayPins[] = {9, 8};
```

```
const int numRelays = sizeof(relayPins) / sizeof(relayPins[0]); // Calculate the number of relays
```

```
void setup()
```

```
{www.sunfounder.com for (int i =
```



```
0; i < numRelays; i++) {  
    pinMode(relayPins[i], OUTPUT); // Initialize each relay pin as an output  
    digitalWrite(relayPins[i], HIGH); // Initially, all relays are turned off (HIGH  
    for low-level trigger)  
}  
  
}  
  
void loop() {  
  
    for (int i = 0; i < numRelays; i++) {  
  
        digitalWrite(relayPins[i], LOW); // Activate current relay (LOW for low-level  
        trigger)  
        delay(1000); // Delay for 1 second  
  
        digitalWrite(relayPins[i], HIGH); // Deactivate current relay  
        delay(2000); // Short delay before act
```

Sensor:

Sony IMX708

Resolution: 11.9 megapixels

Sensor size:

7.4mm sensor

diagonal Pixel size:

1.4 μ m $\times$ 1.4 μ m

Horizontal/vertical:

4608 $\times$ 2592 pixels

Common video modes: 1080p50, 720p100, 480p120

Output:

RAW10

IR cut filter:

Integrated in standard variants; not present in NoIR variants

Autofocus system:

Phase Detection Autofocus

Dimensions:

25 $\times$ 24 $\times$ 11.5mm (12.4mm height for Wide variants)

Ribbon cable length:

200mm

Cable connector:

15 × 1mm FPC

Operating temperature: 0°C to 50°C

Compliance:

FCC 47 CFR Part 15, Subpart B, Class B Digital Device

Electromagnetic Compatibility Directive (EMC) 2014/30/EU

Restriction of Hazardous Substances (RoHS) Directive

2011/65/EU done

- Optional blue pixel color
- Mounting holes: 2
- Dimensions: 30 x 30 mm / 1.2 x 1.2 inch

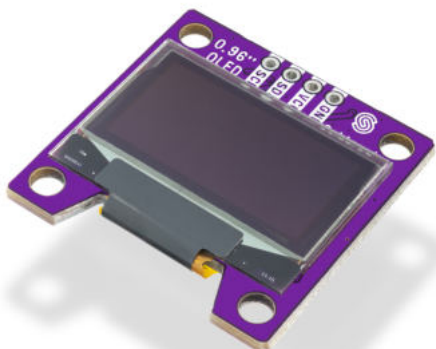
USEFUL LINKS

- [Arduino library](#)
- [Datasheet SSD1315](#)
- [Datasheet SSD1306](#)
- [Open-Source Hardware files](#)

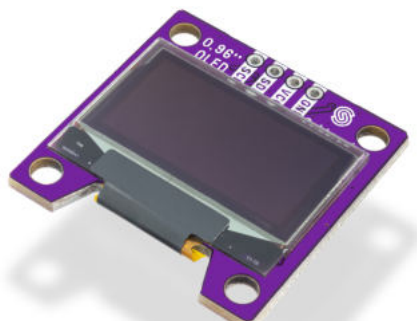
TIPS

If your breakout board is not working, the most common reason is it's not connected properly. Check the pinout and wiring. You'll likely find your culprit there. If everything is properly connected, see if the address of your I2C display is correct before you start programming it. The I2C address of your display is 0x3C. If it doesn't match, be sure to correct it. Make sure the resolution in your program is the same as the resolution of the display. If it's none of that, look through your code for any invalid display instructions. The breakout board works wonderfully with [a Dasduino board](#) due to the easyC system. It is perfect for displaying simple information, like weather or humidity. Connect it to the Internet and get cryptocurrency prices in real-time. We've also added four mounting holes so it won't move an inch once it's mounted. Each pixel on an OLED display is actually a little LED. If you keep it continuously on for over 1000 hours, the screen will start to dim. To prevent that and keep the display bright for a long time, turn off the display when you don't need it on. We've also written a [tutorial](#) for it so you can start using it effortlessly in no time!

OTHER IMAGES



DISPLAY OLED I2C 0.96" SSD1306



Weight	8 g
Color	Blue, White

DESCRIPTION

Looking to add visuals to your Arduino projects? The I2C OLED breakout board is perfect for that. Because of its high contrast, you will be in awe of its crispness! The display is made from hundreds of LEDs, each representing one pixel in a 128x64 grid. Thus, it doesn't need any backlight, and it needs less power than the classic LCD screens. That's 8192 pixels all individually lit at your fingertips!

The design is 5V ready with an onboard regulator, so is compatible with any 3.3V board. The breakout board uses only about 20 mA, depending on active pixels. It is easy to control over the I2C interface. It is even easier to connect with the easyC system - no soldering required!

Display OLED I2C 0.96" SSD1306 options:

The color of the OLED screen on the product can be:

- White
- Blue

FEATURES

- Screen diagonal: 0.96" wide
- Current consumption: ~20 mA (depending on active pixels)
- Logic voltage level: 5V (on I2C header)
- Operating voltage: 5V (onboard regulator for 3.3V)
- Communication: I2C (address: 0x3C)
- Connectors: easyC x2
- Optional white pixel color

Order Code - 2223841.6

Item Name - ESP8266 NodeMCU



Specifications:-

- Microcontroller: Tensilica 32-Bit Risc Cpu Xtensa Lx106
- Operating Voltage: 3.3 V
- Input Voltage: 7-12 V
- Digital I/O Pins (DIO): 16
- Analog Input Pins (ADC): 1
- Uarts: 1
- Spis: 1
- I2cs: 1
- Flash Memory: 4 Mb
- Sram: 64 Kb
- Clock Speed: 80 Mhz
- Usb-Ttl Based On Cp2102 Is Included Onboard, Enabling
- Plug N Play
- Pcb Antenna



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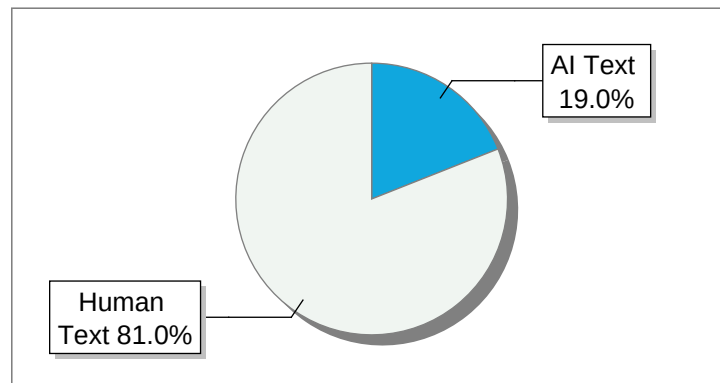
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Paper/Submission ID	4911844
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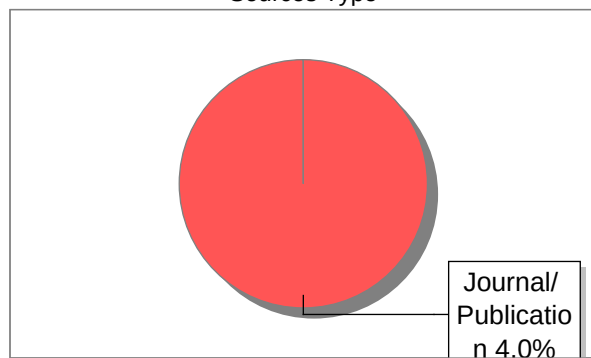
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Title	SMART ASSITIVE SYSTEM FOR ALZHEIMER'S PATIENTS USING IOT AND MACHINE LEARNING
Paper/Submission ID	4911844
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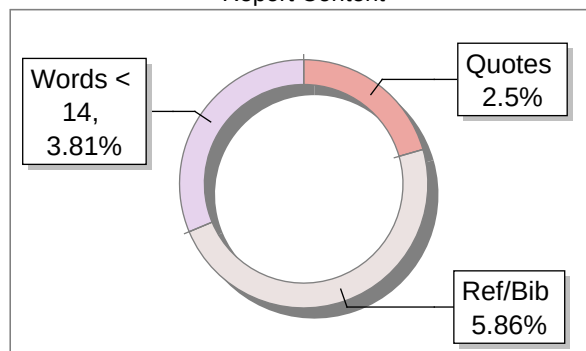
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6	newhorizoncollegeofengineering.in	<1	Publication
7	www.ijarcce.com	<1	Publication